

Problem 13.3.1

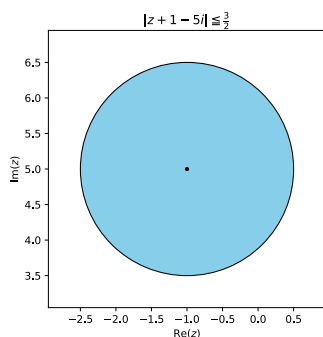
Determine and sketch or graph the sets in the complex plane given by

$$|z + 1 - 5i| \leq \frac{3}{2}$$

Solution

$$|z - (-1 + 5i)| \leq \frac{3}{2}$$

This is a closed disc centered at $(-1 + 5i)$ with radius 1.5



Problem 13.3.3

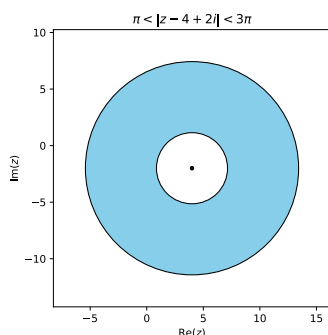
Determine and sketch or graph the sets in the complex plane given by

$$\pi < |z - 4 + 2i| < 3\pi$$

Solution

$$\pi < |z - (4 - 2i)| < 3\pi$$

This is an open disc centered at $(4 - 2i)$ with outer radius 3π , and inner radius π



Problem 13.3.5

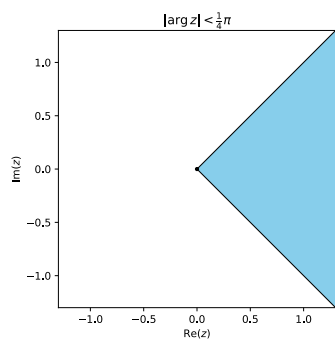
Determine and sketch or graph the sets in the complex plane given by

$$|\arg z| < \frac{1}{4}\pi$$

Solution

$$-\frac{1}{4}\pi < \arg z < \frac{1}{4}\pi$$

This is a wedge from $-\frac{1}{4}\pi$ to $\frac{1}{4}\pi$. There is no bound to the radius of the wedge.



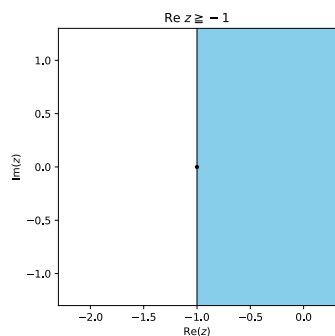
Problem 13.3.7

Determine and sketch or graph the sets in the complex plane given by

$$\operatorname{Re} z \geq -1$$

Solution

This covers the right half of plane bounded at the left by $x = -1$.



Problem 13.3.11

Find $\operatorname{Re} f$, and $\operatorname{Im} f$ and their values at the given point z .

$$f(z) = \frac{1}{1-z} \text{ at } 1-i$$

Solution

$$z = x + iy$$

$$\begin{aligned} f(z) &= \frac{1}{1 - (x + iy)} \\ &= \frac{1}{(1 - x) - iy} \\ &= \frac{1}{(1 - x) - iy} \cdot \frac{(1 - x) + iy}{(1 - x) + iy} \\ &= \frac{(1 - x) + iy}{(1 - x)^2 + y^2} \end{aligned}$$

$$\boxed{\operatorname{Re} f(z) = \frac{1 - x}{(1 - x)^2 + y^2}} \quad \boxed{\operatorname{Im} f(z) = \frac{y}{(1 - x)^2 + y^2}}$$

$$\boxed{\operatorname{Re} f(1 - i) = 0} \quad \boxed{\operatorname{Im} f(1 - i) = -1}$$

Problem 13.3.23

Find the value of the derivative of

$$\frac{z^3}{(z + 1)^3} \text{ at } i$$

Solution

$$f(z) = \left(\frac{z}{z + 1} \right)^3$$

$$\begin{aligned} f'(z) &= 3 \left(\frac{z}{z + 1} \right)^2 \left(\frac{(z + 1) - z}{(z + 1)^2} \right) \\ &= \frac{3z^2}{(z + 1)^4} \end{aligned}$$

$$\begin{aligned} f'(i) &= \frac{3i^2}{(i + 1)^4} \\ &= \boxed{-\frac{3}{4}} \end{aligned}$$

Problem 13.4.3

Is f analytic?

$$f(z) = e^{-2x}(\cos 2y - i \sin 2y)$$

Solution

$$f(z) = u(x, y) + iv(x, y)$$

$$\begin{aligned} u &= e^{-2x} \cos 2y & v &= -e^{-2x} \sin 2y \\ u_x &= -2e^{-2x} \cos 2y & v_x &= 2e^{-2x} \sin 2y \\ u_y &= -2e^{-2x} \sin 2y & v_y &= -2e^{-2x} \cos 2y \end{aligned}$$

Since $u_x = v_y$ and $u_y = -v_x$, f is **analytic**.

Problem 13.4.5

Is f analytic?

$$f(z) = \operatorname{Re}(z^2) - i \operatorname{Im}(z^2)$$

Solution

$$z^2 = (x + iy)^2 = x^2 - y^2 + i2xy$$

$$f(z) = u(x, y) + iv(x, y)$$

$$\begin{aligned} u &= x^2 - y^2 & v &= 2xy \\ u_x &= 2x & v_x &= 2y \\ u_y &= -2y & v_y &= 2x \end{aligned}$$

Since $u_x = v_y$ and $u_y = -v_x$, f is **analytic**.

Problem 13.4.7

Is f analytic?

$$f(z) = \frac{i}{z^8}$$

Solution

$$f'(z) = -8 \frac{i}{z^9}$$

Since f is differentiable everywhere (except $z = 0$), f is **analytic**.

Problem 13.4.13

Is the function harmonic? If yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

$$u = xy$$

Solution

$\Delta u = 0$, and therefore the function is **harmonic**.

$$u_x = y = v_y \quad \implies \quad v = \frac{1}{2}y^2 + h(x)$$

$$v_x = h'(x) = -u_y = -x \quad \implies \quad h(x) = -\frac{1}{2}x^2 + c$$

Therefore,

$$f(z) = xy + i\frac{1}{2}(y^2 - x^2 + c)$$

Problem 13.4.17

Is the function harmonic? If yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

$$v = (2x + 1)y$$

Solution

$\Delta v = 0$, and therefore the function is **harmonic**.

$$v_x = 2y = -u_y \quad \implies \quad u = -y^2 + h(x)$$

$$u_x = h'(x) = v_y = 2x + 1 \quad \implies \quad h(x) = x^2 + x + c$$

Therefore,

$$f(z) = x^2 + x - y^2 + c + i(2xy + y)$$

Problem 13.4.23

Determine a and b so that the given function is harmonic and find a harmonic conjugate.

$$u = ax^3 + bxy$$

Solution

$$\begin{aligned} \Delta u &= u_{xx} + u_{yy} \\ &= 6ax + 0 = 0 \implies a = 0, \text{ } b \text{ is free} \end{aligned}$$

$$u = bxy$$

$$u_x = by = v_y \quad \implies \quad v = \frac{b}{2}y^2 + h(x)$$

$$v_x = h'(x) = -u_y = -bx \implies h(x) = -\frac{b}{2}x^2 + c$$

$$v(x, y) = \frac{b}{2}(y^2 - x^2 + c)$$

Problem 13.5.3

Find e^z in the form $u + iv$ and $|e^z|$ if

$$z = 2\pi i(1 + i)$$

Solution

$$\begin{aligned} z &= 2\pi i(1 + i) \\ &= 2\pi i + 2\pi i^2 \\ &= -2\pi + 2\pi i \end{aligned}$$

$$\begin{aligned} e^z &= e^{-2\pi + 2\pi i} \\ &= e^{-2\pi} e^{2\pi i} \\ &= e^{-2\pi} (\cos 2\pi + i \sin 2\pi) \\ &= e^{-2\pi} \end{aligned}$$

$$u = e^{-2\pi} \quad v = 0$$

$$|e^z| = e^{-2\pi}$$

Problem 13.5.9

Write in exponential form

$$4 + 3i$$

Solution

$$z = re^{i\theta}$$

$$r = \sqrt{4^2 + 3^2} = 5$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right)$$

$$z = 5e^{i \tan^{-1}(3/4)}$$

Problem 13.5.15

Find Re and Im of

$$\exp(z^2)$$

Solution

$$z^2 = x^2 - y^2 + i2xy$$

$$\begin{aligned}\exp(z^2) &= \exp(x^2 - y^2 + i2xy) \\ &= \exp(x^2 - y^2) (\cos 2xy + i \sin 2xy)\end{aligned}$$

$\begin{aligned}\operatorname{Re} \exp(z^2) &= \exp(x^2 - y^2) \cos 2xy \\ \operatorname{Im} \exp(z^2) &= \exp(x^2 - y^2) \sin 2xy\end{aligned}$
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Problem 13.5.17

Find Re and Im of

$$\exp(z^3)$$

Solution

$$z^3 = (x^3 + 3xy^2) + i(3x^2y - y^3)$$

$$\begin{aligned}\exp(z^3) &= \exp((x^3 + 3xy^2) + i(3x^2y - y^3)) \\ &= \exp(x^3 + 3xy^2) (\cos(3x^2y - y^3) + i \sin(3x^2y - y^3))\end{aligned}$$

$\begin{aligned}\operatorname{Re} \exp(z^3) &= \exp(x^3 + 3xy^2) \cos(3x^2y - y^3) \\ \operatorname{Im} \exp(z^3) &= \exp(x^3 + 3xy^2) \sin(3x^2y - y^3)\end{aligned}$
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Problem 13.5.19

Find all solutions and graph some of them in the complex plane.

$$e^z = 1$$

Solution

$$\begin{aligned}e^z &= e^{x+iy} \\ &= e^x (\cos y + i \sin y)\end{aligned}$$

$$e^x (\cos y + i \sin y) = 1$$

$$|e^x| = 1 \quad \implies \quad x = 0$$

$$\arg 1 = 2n\pi \implies y = 2n\pi \quad n \in \mathbb{Z}$$

$$z = 2n\pi i \quad n \in \mathbb{Z}$$

